

Kshitiz Khandelwal

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EDUCATION

BMS Institute of Technology & Management

2024 – 2028

B.E. in Computer Science and Business Systems | CGPA: 9.41

Bangalore, India

- **Relevant Coursework:** Data Structures and Algorithms, Database Systems, Operating Systems, Programming in C/C++, Internet of Things

Narayana PU College – Class XII, 85% (CBSE)

2022 – 2024

BGS World School – Class X, 94% (ICSE)

2020 – 2022

TECHNICAL SKILLS

Programming Languages : Python, C, C++, SQL, HTML/CSS

Backend & APIs : Flask, FastAPI, REST API Design, SQLite, Server-side Scripting

Data & ML : scikit-learn, LightGBM, XGBoost, Deep Learning, SHAP, pandas, NumPy

Embedded & IoT : Zephyr RTOS, Raspberry Pi Pico (RP2350), GPIO, UART, I2C, MicroPython, C for microcontrollers

Tools : Git, Linux, Streamlit, Jupyter Notebook, VS Code

PROJECTS

Atlas Identity OS – Local-First AI Personal Assistant | Rust, Tauri, React, TypeScript, SQLCipher, ONNX | [GitHub](#)

- Built a fully local, privacy-first AI desktop assistant (Windows) using **Rust + Tauri v2** with a React/TypeScript frontend; zero cloud dependency – all LLM inference, voice transcription, and vector search runs on-device via Ollama.
- Designed a **26-question Behavioral Evidence Engine** (Situational Judgment Tests, forced-choice MCQs) grounded in OCEAN/MBTI personality research to build a behaviorally accurate user persona fingerprint injected into every AI conversation.
- Implemented **AES-256 encrypted SQLite (SQLCipher)** vault with Argon2id key derivation (64 MB memory-hard, GPU-resistant), BIP39 recovery phrase generation, and in-memory key zeroization via the secrecy crate.
- Built an **agentic OS control engine** (13 tools) enabling natural language commands to play YouTube/media, launch apps, open websites, lock/sleep the PC, query live CPU/RAM diagnostics via WMI PowerShell, set timers, and capture screenshots.
- Integrated local **ONNX vector embeddings** (bge-small-en-v1.5, 384-dim) stored in sqlite-vec for KNN semantic memory retrieval; implemented hybrid retrieval scorer combining cosine similarity, graph centrality, and recency decay.

Finova – AI Personal Finance System | Python, IBM Granite LLM, SQL

- Built a personal finance app with SQL modules for expense tracking, budgeting and investment planning; connected IBM Granite AI via API for personalized financial recommendations.
- Led a small hackathon team – split up the work and kept the build on track to finish within the deadline.

Krishi360 – AI Agriculture Platform | Python, Machine Learning, REST API | [Live Demo](#)

- Built a REST API-connected platform that gives farmers crop recommendations based on soil data, rainfall, market trends and disease risk; implemented the backend logic for the Crop Suitability Index (CSI).
- Led the team at Smart India Hackathon – decided who builds what, kept things moving and presented the final product to judges; won **3rd Place in the Ideathon**.

ISRO BAH 2026 – Predictive NOC Copilot | Python, XGBoost, Prophet, ChromaDB

- Designed the full system layout for **ISRO Build-A-Hackathon 2026 (PS13)** and wrote documentation covering each module's purpose and data flow.

TON-IoT Intrusion Detection System | Python, LightGBM, XGBoost, scikit-learn, SHAP

- Built an end-to-end ML pipeline on the TON_IoT dataset (211K network traffic records): handled missing values, log-transformed skewed features, label-encoded categorical fields, and engineered protocol/state-based features.
- Trained and compared 4 classifiers (Random Forest, LightGBM, XGBoost, MLP) for both binary (benign/attack) and 10-class attack-type detection; organized results in a structured comparison table.

- Applied SHAP analysis to identify the most influential features per model, providing interpretable insights into network-level attack patterns across IoT traffic.

ECG Hybrid Classification System | Python, EfficientNet-B4, LightGBM, Deep Learning | [Live Demo](#)

- Built a two-stage classification pipeline on the PhysioNet/CinC 2020 ECG dataset: EfficientNet-B4 for deep feature extraction from 12-lead ECG signals, followed by LightGBM for final classification – achieving **92% accuracy** and **97.1% weighted AUC**.
- Designed a patient risk-scoring layer that maps model confidence to a clinical risk percentage, making predictions interpretable for non-technical users.
- Deployed results to an interactive dashboard (Next.js on Vercel) displaying per-class predictions, confidence scores, and risk levels across 12-lead ECG recordings.

EXPERIENCE

Research Intern | [Certificate](#)

Mar 2026 – May 2026

Indian Institute of Information Technology Tiruchirappalli (IIIT Trichy)

Remote

- Completed a research internship under the **AICTE-QIP-PG Certificate Programme on Machine Learning and Cyber Physical Systems**, supervised by Dr. R. Jennie Bharathi (Asst. Professor/CSBS, BMSIT&M).
- Contributed to the research project “*An SNN-based Split Federated Learning Framework for Privacy Preserving Intrusion Detection in IoMT Networks*” – focusing on intrusion detection over healthcare IoT data using federated/split learning architectures and spiking neural networks (SNNs).
- Applied privacy-preserving machine learning techniques on IoMT network traffic data, relevant to backend data pipeline and distributed system design.

IoT & Edge AI Outreach Workshops, BMSIT&M | [Event Photos](#)

Co-Organizer & Instructor

SCJIT & Vivekananda Institute of Technology

- Conducted hands-on IoT and Edge AI workshops at two colleges – **SCJIT** and **Vivekananda Institute of Technology** – covering embedded systems, Raspberry Pi Pico, and real-time sensor interfacing.
- Co-planned logistics, prepared session material and adjusted technical depth based on participants' backgrounds; handled on-day coordination and collected post-session feedback.

IIC Council, BMSIT&M

Nov 2024 – Apr 2025

Marketing Team Member

- Handled promotions and outreach for college innovation events; cold-called businesses to get sponsors for a national-level Ideathon.

Talents Day Cultural Event, BMSIT&M

Event Organizer

- Helped organize a college cultural event (music, art and craft); handled scheduling, promotions and on-day coordination.

ACHIEVEMENTS

- **Research Internship** – IIIT Tiruchirappalli (AICTE-QIP-PG Programme), on SNN-based split federated learning for privacy-preserving intrusion detection in IoMT networks (Mar–May 2026).
- **ISRO Build-A-Hackathon 2026** – Submitted a solution for PS13 (Predictive NOC Copilot for Secure MPLS Operations).
- **3rd Place** – Ideathon, Smart India Hackathon, for the Krishi360 AI Agriculture Platform.
- Academic **CGPA: 9.41** in B.E. Computer Science and Business Systems at BMSIT&M.
- Delivered multiple end-to-end software projects as team lead through self-learning, covering backend development, APIs, databases and system design.